

Education

2022-2026 Research in Mechanical Engineering (Ph.D.) Imperial College London

Topic: Erosion During Impact of Droplets on Wind Turbine Blade Surfaces Full-Scholarship

Industrial Collaborations:

- International Energy Agency (IEA) Wind TCP Task 46
- Offshore Renewable Energy (ORE) Catapult, UK
- Equinor ASA, Norway

Research Publications:

1. Refereed Journal Publications

- **H. Hao***, M. Charalambides, Y. Hardalupas, A. Sergis, A. Taylor (2026). Analytical solution of droplet impact on solid surfaces at short and intermediate times. *J. Fluid Mech.*, **in press**.
- **H. Hao**, H. Chen, S. Xu, M. Charalambides, A. Sergis, A. Taylor, H. Chen, Bing Wang*, Y. Hardalupas* (2026). Hydrodynamic and material response to droplet impact on coated turbine blades: Implications for cavitation erosion. *J. Fluid Mech.*, **in press**.
- **H. Hao***, L. Domenech, F. Sánchez* (2026). Modelling rain erosion surface damage initiation in turbine blades based on inspection data at wind farms. *Results Eng.*, **in press**.
- F. Sánchez*, **H. Hao***, L. Domenech, Y. Hardalupas, K. Dyer, V. García-Peñas, M. Charalambides, M. Ibáñez-Arnal, A. Sergis, A. Taylor (2026). Rain erosion damage initiation assessment of wind turbine blades leading edge protection systems based on laboratory testing data: a review and case study. *Results Eng.*, **in press**.

2. Manuscripts Under Review

- **H. Hao***, M. Charalambides, Y. Hardalupas, A. Sergis, A. Taylor (2026). An analytical-numerical coupled model of liquid droplet impact on solid material surfaces. *J. Fluid Mech.*, **under review**.

3. Manuscripts Ready for Submission

- **H. Hao**, H. Chen, M. Charalambides, A. Sergis, A. Taylor, Bing Wang*, Y. Hardalupas*. Impact-induced cavitation dynamics in non-circular cylindrical droplets: Shape and orientation effects. Target journal: *Phys. Rev. Fluids*.
- **H. Hao***, A. Sergis, A. Taylor, Y. Hardalupas, M. Charalambides. Cavitation effects in droplet impact erosion: Analytical fatigue modelling with an extended Springer model and numerical damage simulation. Target journal: *Wear*.
- **H. Hao***, A. Sergis, A. Taylor, Y. Hardalupas, M. Charalambides. Evaluation of the performance of an analytical-numerical coupled method for droplet impacts on soft material surfaces. Target journal: *Wear*.
- **H. Hao**, O. Ghita*, Y. Hardalupas*. On development of thermoplastic LEPs and their mechanical performance. Target journal: *Wear*.
- **H. Hao***, M. Charalambides, A. Sergis, A. Taylor, Y. Hardalupas. Influence of droplet morphology on fluid dynamics and impact erosion in renewable energy systems. Target journal: *J. Fluids Struct.*

4. Technical Reports

- **H. Hao**, L. Domenech, F. Sánchez (2025). Use of in-field turbine blades inspection data for the modelling of the rain erosion damage initiation. IEA Wind TCP Task 46 Technical Report (underpinning journal publication in *Results Eng.*).
- F. Sánchez, **H. Hao** (2025). Review and assessment of erosion damage models based on fundamental material properties and DNV-GL-RP0573. IEA Wind TCP Task 46 Technical Report (underpinning journal publication in *Results Eng.*).

2020-2021 Advanced Mechanical Engineering (M.Sc.)

Imperial College London

Distinction (Ranked within the top few)

Full-Scholarship

2019-2019 **Exchange Student** **Massachusetts Institute of Technology**
Exchange programme between ICL and MIT Full-Funding
2017-2020 **Mathematics with Applied Mathematics (B.Sc.)** **Imperial College London**
First class honours (Ranked within the top few)

Research Experience & Collaborations

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- 2026.4-Present** **Research Associate, Imperial College, UK & Tsinghua Uni. CN**
- Developed fluid-structure interaction model with phase changes.
- 2022.9-Present** **Participant, International Energy Agency (IEA) Wind TCP Task 46**
- Developed erosion lifetime prediction tools; work contributed to wind industrial practice.
- 2023.2-2023.4** **Research Visitor, CEU Cardenal Herrera University, Spain**
- Validated state-of-art wind-blade erosion models; work contributed to IEA Wind TCP Task 46.
- 2020.10-2021.9** **MSc Project, Imperial College, UK & Equinor ASA, Norway**
- Developed single droplet impact models; work contributed to industry on wind-blade erosion.
- 2019.10-2020.9** **Self-driven Project, Imperial College & Uni. of Oxford, UK**
- Developed numerical simulations on wave chaotic between oscillating wall and porous layers.
- 2019.6-2019.8** **International Research Opportunities Programme (IROP), MIT, USA**
- Developed and benchmarked Julia ODE solvers; results presented at the 2019 JuliaCon.

Awards & Scholarships

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- Awarded a full PhD scholarship at Imperial College London.
 - Awarded a full Equinor MSc scholarship at Imperial College London.
 - Awarded a full funding at Imperial College London for IROP exchange programme.
 - Honoured Unwin Postgraduate Prize at Imperial College London.
 - Honoured Dean's List at Imperial College London.
 - Honoured the G-research prize at Imperial College London.
 - Honoured the Tensor Society Mathematics Prize at Imperial College London.
 - Honoured Imperial College Trust fundings for international academic conferences.
 - Honoured Old Centralian's Trust fundings for international academic conferences.

Teaching

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- 2024.9-2025.6** **Co-supervision to FYP Project, Imperial College Dept.MechEng, UK**
- 4th year undergraduate, on topics of numerical simulations of oblique droplet impacts.
- 2024.7-2024.9** **Co-supervision to UROP Project, Imperial College Dept.MechEng, UK**
- 2nd year undergraduate, on topics of droplet impacts in combustion engine.
- 2023.9-2024.6** **Co-supervision to FYP Project, Imperial College Dept.MechEng, UK**
- 4th year undergraduate, on topics of mooring line system of floating offshore wind turbine.
- 2023.9-2024.6** **Co-supervision to FYP Project, Imperial College Dept.MechEng, UK**
- 4th year undergraduate, on topics of Navier-Stokes equation and flow fields analysis.
- 2023.9-2024.1** **Co-supervision to LRP Project, Imperial College Dept.MechEng, UK**
- 3rd year undergraduate, on topics of leading-edge erosion on wind turbine blades.
- 2022.9-2023.6** **Co-supervision to FYP Project, Imperial College Dept.MechEng, UK**
- 4th year undergraduate, on topics of numerical simulations of multiple droplet impacts.
- 2022-Present** **Graduate Teaching Assistant, Imperial College Dept.MechEng, UK**
- 2019-2020** **Undergraduate Teaching Assistant, Imperial College Dept.MechEng, UK**
- 2025.10-Present** **Maths Mentor, Tutorfair Foundation, UK**
- 2018.7-2018.8** **Volunteering Teaching Assistant, Gansu Province, China**